QpiAI <> Q-BITS, Nitte Meenakshi Institute of Technology

QUANT-A-MAZE 3.0

INDUSTRY PROBLEM STATEMENTS AND JUDGEMENT RUBRICS

1. PURPOSE AND NON-NEGOTIABLE DESIGN PRINCIPLES

The four tracks are considered finalized. Participants' submitted presentations are indicators of interests and technical directions; they are not finalized challenge statements and should not be copied as such. The purpose of this brief is to translate those interests into problems recognizable as useful, consequential problem statements by professionals in relevant industrial domains, viz. logistics, energy, defense, finance, pharmaceuticals, health systems, security, agriculture and industrial R&D.

Problem statements and associated deliverables are open to interpretation and are intended to provide a foundational direction rather than impose rigid constraints. Participants are encouraged to exercise creative freedom and explore innovative approaches, provided their solutions preserve and build upon the core intent and objectives outlined. Participant's chosen problem statement and their solution must have: (1) a suitably attributable industrial domain; (2) an observable business or scientific outcome; (3) a mathematically valid quantum/optimization or track-specific technical component; and (4) an end-to-end implementation without replacing the real problem with an unrelated toy.

There are five problem statements per track: Quantum Machine Learning (QML), Post-Quantum Cryptography (PQC), Web3 & Blockchain, and Generative AI & Machine Learning. Teams should select freely within their confirmed track.

2. TECHNOLOGY ROLE: QpiAI QCloud & QpiAI Opt

QpiAI QCloud / Quantum SDK is for actual quantum-circuit construction, measurement, circuit feature representations and experiments on an available simulator or backend.

- Install the open source QpiAI Quantum SDK:
  > github.com/qpiai/quantum-sdk ; `pip install qpiai-quantum`
- Register and get free API key:
  > qcloud.qpiai.tech

QpiAI Opt / Opt SDK is for substantive, correctly formulated combinatorial decisions, especially QUBO / Ising and native-higher order HUBO.

- Install the open source QpiAI Opt SDK:
    > github.com/qpiai/qpiai-opt ; `pip install qpiai-opt[quantum]`
- Register and get free API key:
    > opt.qpiai.tech

Every QML challenge **REQUIRES** meaningful use of **BOTH** QpiAI QCloud and QpiAI Opt. In other tracks, QpiAI Opt **MUST** be used for the real scheduling, selection or allocation task wherever applicable. Similarly, QpiAI QCloud **MUST** be used meaningfully wherever applicable. Consequently, usage of QpiAI technologies is incentivized in the judgement rubrics (Section 8).

3. THEMATIC ALIGNMENT AS PER PARTICIPANT SUBMISSIONS

QML domains: fleet routing, EV charging, smart grids, battery health and second life, molecule selection, financial risk, recovery logistics and quantum feature representations.

PQC domains: cryptographic inventories, migration, secure messaging, quantum-resistant signatures, satellite commands, embedded gateways and formalized verification.

Web3 domains: crop provenance, payments, evidence custody, verified records, language-data rights, accountable agents, traceability and revocation.

GenAI/ML domains: incident triage, city buses, healthcare supplies, agronomy, retail inventory, document dependencies, safety and operating decisions.

4. TRACK I: QUANTUM MACHINE LEARNING

Mandatory for ALL five: QpiAI QCloud Quantum + QpiAI Opt QUBO / Ising / HUBO optimization.

QML-01: Delay-Risk-Aware Fleet Dispatch for Contract Logistics

Problem. A contract logistics operator has finite vehicles, vehicle-specific capacities, delivery windows and contractual commitments. Traffic and service-time uncertainty can make a cheap-looking route fail its SLA. Produce vehicle routes that are operationally feasible and minimize actual transport/labor expense together with estimated service-failure exposure.

Deliver. Exact vehicle-to-route plan, ordered stops, ETA/risk for each service commitment and total financial exposure. Evaluate SLA violations, valid coverage, cost, prediction calibration, solver quality and runtime.

QML-02: Grid-Constrained Commercial EV Fleet Charging

Problem. Vehicle charging must finish before real departure deadlines without exceeding feeder/transformer limits or driving expensive power peaks. Select charger-and-time allocations minimizing energy tariff costs and demand charges while meeting all confirmed departure requirements.

Deliver. Vehicle/charger/time-slot plan, energy delivered, predicted departures, aggregate kW loading and bill. Measure departure compliance, peak kW, tariff cost and capacity violations.

QML-03: Retired Battery Qualification and Second-Life Allocation

Problem. Retired batteries differ in health, electrical properties and degradation. Safely decide which units qualify for reuse, grouping or recycling while protecting minimum compatibility and value constraints.

Deliver. Unit-by-unit reuse/testing/recycling disposition, grouped capacity, compatibility rationale and uncertainty. Evaluate health estimation error, forbidden-assignment count, value, pack mismatch.

QML-04: Assay-Constrained Selection of Drug-Discovery Compounds

Problem. A drug-discovery team can test only a limited number of molecules and must trade predicted activity against chemical novelty, diversity, assay expense and availability. Select a high-value next experimental batch for one specified target and assay.

Deliver. Specific molecules, source IDs, cost, predicted measurements/uncertainty, scaffold diversity and validation plan. Evaluate retrospective enrichment on held-out assay results, novel scaffold coverage and selection value.

QML-05: Discrete Foreign-Exchange Hedge Allocation

Problem. Dated contractual foreign-currency cashflows expose a company to adverse FX moves, but forwards/options are purchasable only in discrete tradable lots and constrained by approved risk policy, credit and liquidity. Select economically appropriate hedge lots and dates.

Deliver. Hedge tickets, quantities, maturities, quoted costs and residual scenario exposures with compliance report. Benchmark held-out risk and implementation cost.

5. TRACK II: POST-QUANTUM CRYPTOGRAPHY

PQC-01: Verification-Gated Cryptographic Migration Across Banking Applications

Problem. A bank has vulnerable RSA/ECC-dependent services, certificates and software dependencies that cannot all transition simultaneously without interrupting payment services. Identify current cryptographic exposure and prioritize a provably compatible migration schedule.

Deliver. Inventory with file/service evidence, migration sequence, working before-and-after protocol traces, compatibility tests and rollback plan. Evaluate detection precision/recall on labelled code and configs, protocol correctness, overhead and feasible migration dependencies.

PQC-02: Post-Quantum Authentication of Automotive Firmware Updates

Problem. Software-defined vehicles receive firmware updates on constrained ECUs. Forged/rolled-back firmware must be rejected, while real signature sizes, verification time and recovery constraints have to remain acceptable over long platform lives.

Deliver. Runnable signer and verifier, signed update manifest, measured signature bytes/latency/RAM/flash, failure-recovery results and negative-test suite. Hardware profiles must be measured on an identified available device.

PQC-03: Quantum-Resistant Satellite Command Authentication

Problem. Satellites accept high-consequence commands over limited-bandwidth communications, often with low repairability. Unauthorized or replayed instructions must be rejected without exhausting onboard computation or link budgets.

Deliver. Ground-side signing, onboard-side receiving verification, accepted/rejected log and measured resource budget. Test tampered, replayed, expired, unauthorized and forged commands.

PQC-04: Long-Term Confidentiality in Hospital-to-Laboratory Data Exchange

Problem. Medical reports are retained for long periods; captured encrypted traffic may face future quantum threats. Migration to new public-key mechanisms must protect confidential transfers without breaking interoperability.

Deliver. Complete secure record exchange, validated protocol handshake, negative-case tests and an explicit key-management/compatibility report. Evaluate session establishment correctness, authorization rejection, data integrity and end-to-end latency.

PQC-05: PQC Deployment in Resource-Constrained Industrial Gateways

Problem. Industrial gateways may remain operational for a decade or more under severe processor, memory, frame-size and downtime constraints. Cryptographic migration must maintain authenticated communication and configuration updates without interrupting control functions.

Deliver. Working secure gateway session, signed-update verification, hardware/resource measurements and deployment plan. Distinguish device measurements from measurements on a desktop emulator.

6. TRACK III: WEB3 & BLOCKCHAIN

WEB3-01: Agricultural Procurement, Quality Grading and Payment Settlement

Problem. Agricultural lot value depends on verified delivered weight, quality grading, loss, transfer and contract terms. Unverifiable deductions and duplicated claims create disputes between growers, collectors, warehouses and processors.

Deliver. Full lot ancestry, mass/quantity reconciliation, attested grading, settlement records, dispute evidence and feasible commercial allocation. Evaluate false/duplicate settlement rejection, quantity consistency, contractual fulfillment and cost.

WEB3-02: E-Waste Collection, Recovery Yields and Auditable Material Claims

Problem. E-waste collected from disparate channels can be misreported, lost between handovers or counted twice toward material recovery. An industrial recycler needs traceable custody and reconcilable physical yield evidence.

Deliver. Traceable batch graph, processor evidence, validated mass-balance, duplicate-claim checks and dispatch assignment. Evaluate independent inventory reconciliation, missing handovers, cost and licensed-facility compliance.

WEB3-03: Cross-Enterprise Digital Passports for EV Battery Reuse and Recycling

Problem. Batteries change ownership and use across manufacturers, repairers, remanufacturers and recyclers. Every party needs authenticated identity and condition history but should only access information authorized for its role.

Deliver. Verified battery-lifecycle state, independently inspectable custody/history evidence, access-control tests and disposition decisions. Measure unauthorized write rejection, lineage continuity, missing attestation detection and allocation feasibility.

WEB3-04: Rights-Gated Regional-Language Dataset Licensing for AI Training

Problem. AI model training datasets may carry different rights for commercial training, geography, derivative works, redistribution, expiry or revocation. Unauthorized data ingestion creates substantial contractual and reputational exposure.

Deliver. Signed rights ledger, ingestion decision API, accepted/rejected corpus manifests, audit log, licensing-cost allocation and adversarial policy checks. Evaluate false authorizations, correct denial behavior, traceability and coverage.

WEB3-05: Policy-Bound Autonomous Procurement and Machine-to-Machine Payments

Problem. Autonomous software can buy low-value supplies/services efficiently but must not exceed delegated authority, buy from unauthorized vendors or obey malicious instructions that manipulate payment policy.

Deliver. Agent request, approved purchase basket, signed authority, executable payment on a controlled test environment, transaction proofs and revocation. Evaluate unauthorized-payment rejection, purchasing costs, budget enforcement and verifier failure handling.

7. TRACK IV: GENERATIVE AI & MACHINE LEARNING

GENAI-01: Freight Exception Resolution Before Demurrage and Delivery Penalties

Problem. Documentation, customs release, terminal deadlines and truck bookings are fragmented across carrier emails, shipping documents, status feeds and internal notes. Missing an action can cause avoidable demurrage, detention, storage or missed deliveries.

Deliver. Container-level blocker, verified evidence, time-bound responsible action, feasible truck/slot allocation and projected avoidable costs. Assess evidence precision, missed blockers, false escalations, deadline compliance and historical expense.

GENAI-02: Security Incident Triage and Qualified Analyst Assignment

Problem. SOC analysts must group related identity, endpoint and cloud alerts, preserve evidential context and assign urgent cases to qualified responders under capacity and response-time contracts.

Deliver. Auditable incident queue, links to contributing logs, assigned specialists, response plans and uncertainty. Evaluate evidence accuracy, grouping precision, important incident recall, response SLA adherence and analyst workload.

GENAI-03: Bus Headway Recovery and Passenger Disruption Updates

Problem. Traffic and uneven boarding cause buses to bunch. Dispatchers choose whether to hold, short-turn, release a spare or make no change without breaking driver, stop-coverage and fleet constraints.

Deliver. Specific dispatcher hold/short-turn/spare orders, predicted revised arrivals and accurate bilingual or multilingual passenger information where relevant. Evaluate predicted ETA error, headway regularity, missed service, feasibility and notification truthfulness.

GENAI-04: Expiry-Aware Essential Medicine Redistribution Across Health Facilities

Problem. Some medical facilities stock out of essential drugs while nearby facilities hold eligible lots close to expiry. Redistribution is constrained by actual inventories, drug identity, minimum stock reserve, shelf life, permitted transport and cold chain.

Deliver. Auditable lot-level transfer orders, quantities, source/destination, temperature requirements, shortage reduction and expected expiry avoidance. Evaluate actual stockout reductions, expiry and inventory feasibility.

GENAI-05: Plant-Disease Investigation Coupled to Real Agronomy-Service Capacity

Problem. Visually similar crop damage may arise from pests, disease, nutrient stress or weather. Field managers need a verified diagnosis pathway and realistic scheduling of

agronomists, testing services and approved supplies; generic crop-image diagnosis is insufficient.

Deliver. Auditable per-field investigation plan, validated next test, appointment, available permitted inputs (when independently confirmed) and service orders. Evaluate diagnosis quality on independent field observations, unsafe advice, turnaround and feasibility.

8. JUDGEMENT RUBRICS: QpiAI TECHNOLOGY METRIC

The following scoring criteria shall be used for scoring the use of QpiAI technologies in the submissions, and shall be incorporated into the judgement rubrics accordingly. The 20 points score assumes a 100 points-based scoring system; the scaling shall be adjusted as applicable.

A. USE OF QPIAI TECHNOLOGIES — 20 points.

Judges must score ACTUAL invocation, relevance, correctness and measured integration of QpiAI technology, not mentions in slides or a decorative circuit.

A1. Correct, meaningful QpiAI product choice and domain integration — 5 points.

Does the selected QpiAI component address a genuine computation or domain decision? Are inputs and outputs materially connected to the validated business outcome?

A2. Technical formulation and correct use of features — 10 points.

QpiAI QCloud and associated technical correctness — 5 points (50%).
QpiAI Opt and associated technical correctness — 5 points (50%).

A3. Working QpiAI implementation with recorded execution evidence — 5 points.

Working SDK and solver runs, recorded outputs and reproducibility.